Thin-walled MSG shell vs. 3D solid: benchmark summary

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What was tested. The MSG thin-walled shell cross-sectional analysis, in Reissner–Mindlin (RM, C^0) and Kirchhoff (KF, C^1) wall kinematics, was compared term-by-term against a 3D solid (FEniCSx, VABS-equivalent) on four sections—a closed circular **tube** and an open flat **strip**, each **isotropic** ($E = 70\,\mathrm{GPa}$, $\nu = 0.3$) and anisotropic [45/−45] (E_1=37, E_2=E_3=9, G=4 GPa)—over a thin-to-thick sweep $h/R = h/W = 0.01$–0.20, referenced at the wall mid-surface (centre) and outer mould line (OML). Stiffnesses use both the matrix index and the engineering name: C_{11}=EA, C_{22}=GA_2, C_{33}=GA_3, C_{44}=GJ, C_{55}=EI_2, C_{66}=EI_3.

Key results.

- **Tube (closed):** both models match the solid to $< 1\%$ on the full 6×6 at the centre, thin to thick; OML adds an $O(h/R)$ drift.
- **Strip EA, EI_3:** exact ($< 2\%$) for both models.
- **Strip torsion GJ:** the open section exposed (and we corrected) the RM twist measure to the full $-2\kappa_1$; RM now recovers St-Venant *exactly* and *beats* Kirchhoff for the thick wall (h/W=0.20: RM −0.1% vs. KF +22%), while the tube GJ is preserved.
- **Strip weak bending EI_2:** captured at the centre (KF a few %; RM slightly soft from bend–twist).
- **Strip transverse shear GA:** method-sensitive. GA_3 (thickness direction of a wide strip) is a 2D, h^3-governed field no 1D-shell resolves (both models err); in-plane GA_2 is exact (iso) and carried by KF but under-predicted by RM for [45/−45].
- **Couplings:** C_{15}, C_{16}, C_{56} are *zero at the centroid* for these symmetric sections (C_{15} appears only as an OML parallel-axis artifact). The genuine [45/−45] couplings are extension–twist C_{14} and shear–bending C_{25}/C_{36} (large for the open strip, an order smaller for the closed tube).

Bottom line. Use the mid-surface reference. EA, EI and (with the twist correction) GJ are trustworthy in both models for open and closed sections; RM's advantage is thick-section torsion. The open-section transverse shear (GA) is the remaining method-sensitive quantity. Full tables, all four error plots, and the geometry/orientation figures are in the detailed report (`benchmark_report.pdf`).

Table 1: Centre-reference shell-vs-solid % error of the principal Timoshenko stiffnesses at the thickest section ($h/R = h/W = 0.20$). C_{33} (GA_3) of the strip is the 1D-shell-limited thickness-direction shear (large error expected for both models); C_{44} (GJ) shows RM beating KF for the thick open strip.

case	model	C_{11} (EA)	C_{22} (GA_2)	C_{33} (GA_3)	C_{44} (GJ)	C_{55} (EI_2)	C_{66} (EI_3)
Isotropic tube	RM	+0.0	-0.8	-0.8	-0.3	-0.5	-0.5
	KF	+0.0	-1.1	-1.1	+0.3	-0.6	-0.6
Anisotropic $[45/-45]$ tube	RM	-0.1	-0.7	-0.7	-0.2	-0.2	-0.2
	KF	+0.1	-1.3	-1.3	-0.5	-0.6	-0.6
Isotropic strip	RM	-0.0	-0.0	+134.9	-0.1	-0.1	-0.0
	KF	-0.0	-0.0	+57.0	+14.4	-0.1	-0.0
Anisotropic $[45/-45]$ strip	RM	+0.3	-32.9	+53.7	+0.2	-10.6	+0.8
	KF	+1.8	+7.4	-11.2	+22.3	+6.6	+4.7

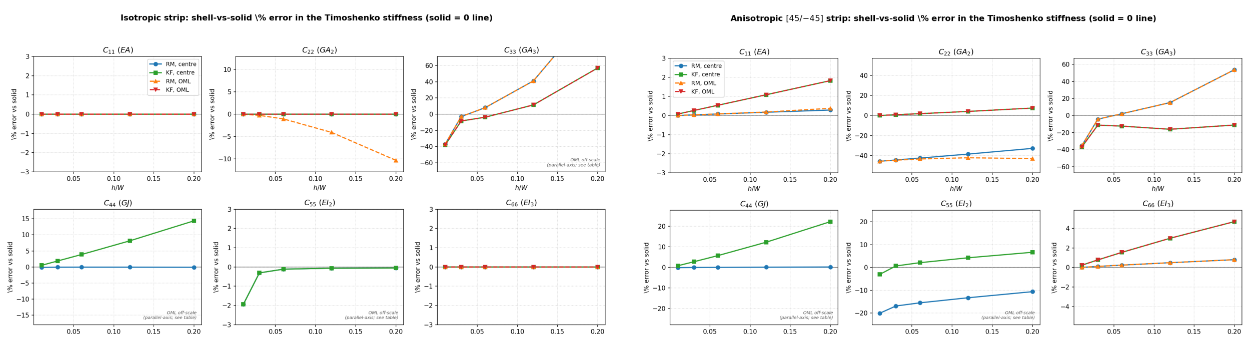

Figure 1: Strip % error vs. the solid (centre = solid lines, OML = dashed). Note C_{44} (GJ): RM flat at zero, KF rising—RM is more accurate for thick open-section torsion. OML EI_2/GJ are off-scale reference offsets.

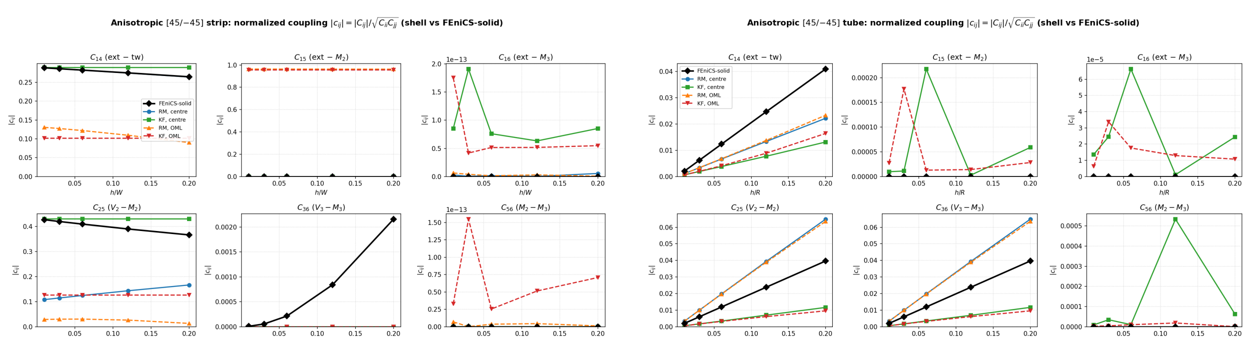

Figure 2: Normalized couplings $|c_{ij}|$ for the $[45/-45]$ strip (left) and tube (right). C_{14}, C_{25} are the genuine material couplings; $C_{15}, C_{16}, C_{56} \approx 0$ at the centre.